Configure Weighted Routing Policy in Amazon Route 53

Summary:-

In this lab, you will configure a **Weighted Routing Policy** in Amazon Route 53 to distribute DNS traffic across multiple EC2 web servers. You will create weighted A records for two web servers, verify traffic distribution using DNS queries, understand how record weights affect traffic allocation, and learn how Route 53 Health Checks improve high availability by routing traffic only to healthy resources.

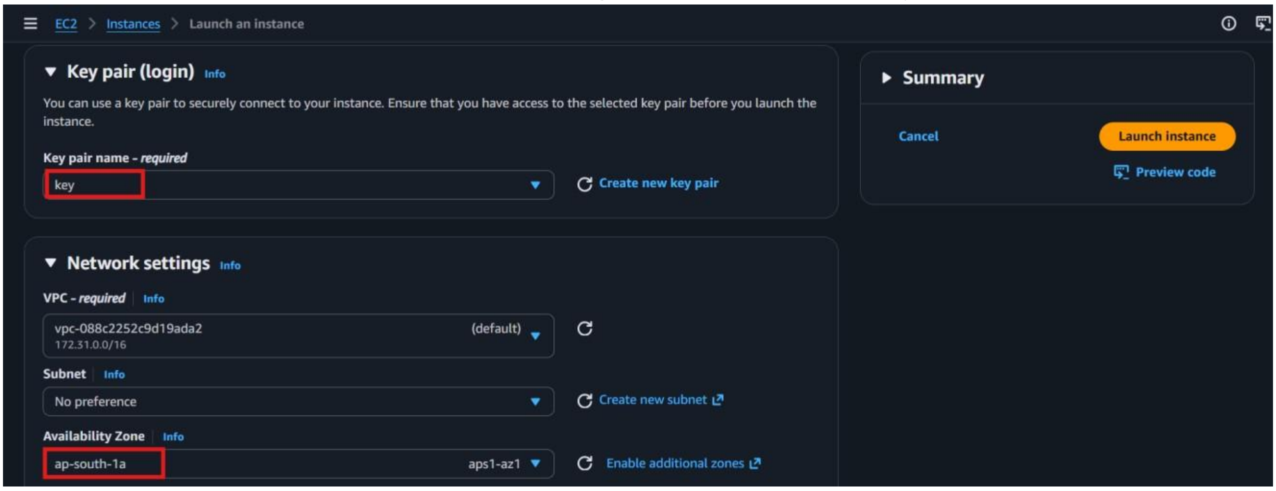
Let's Start:-

- Configure a Weighted Routing Policy in Amazon Route 53.
- Create multiple weighted A records for the same domain name.
- Verify DNS traffic distribution between multiple web servers.
- Understand how record weights determine traffic allocation.
- Learn the importance of Route 53 Health Checks for high availability.

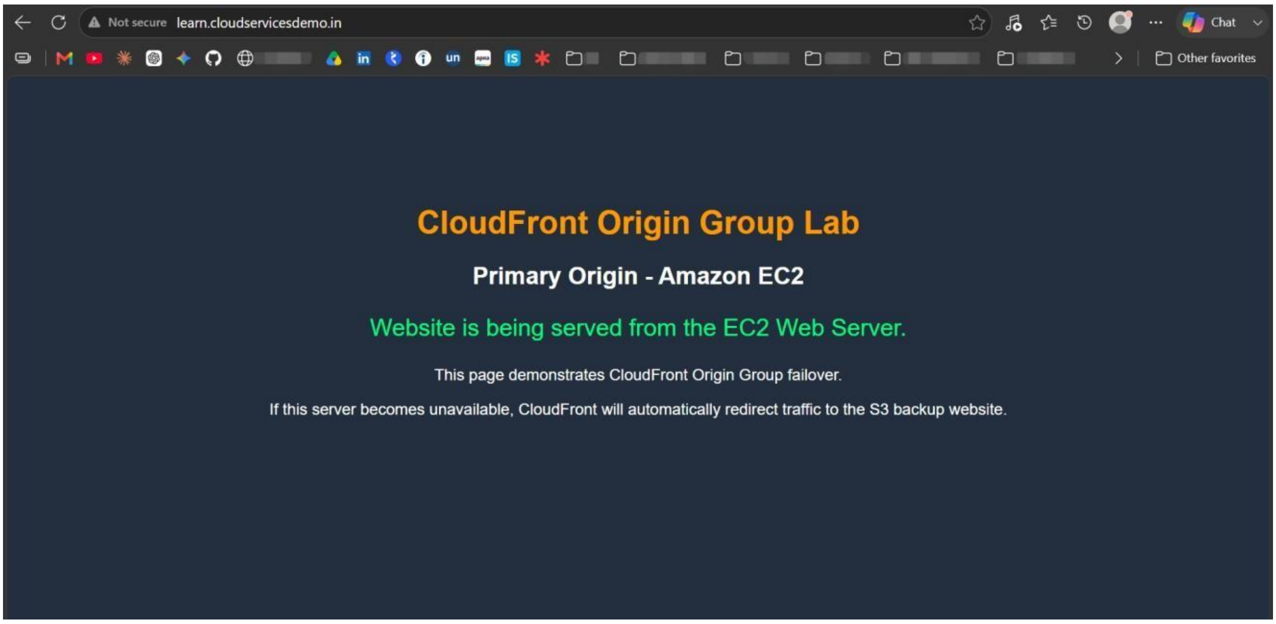
Lab Steps:-

Step 1 – Verify the Web Servers

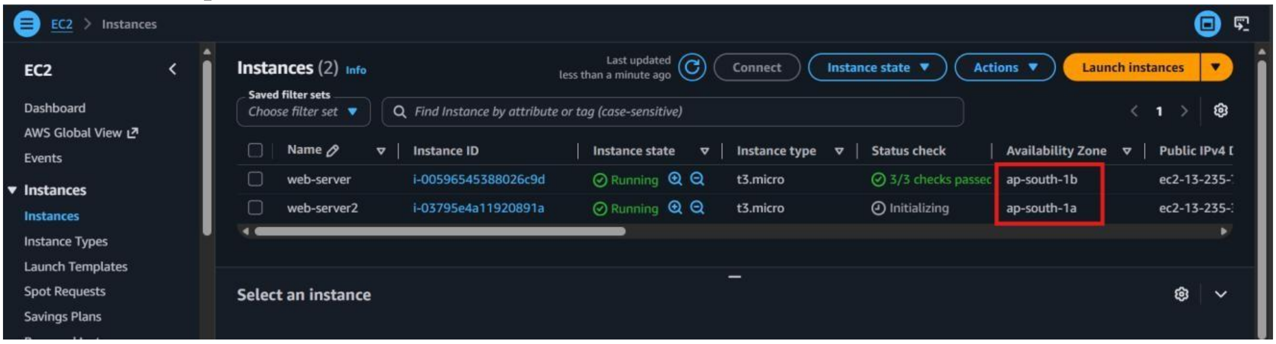
- **Note:** Before attempting this lab, complete **Steps 1–2** of the **Route53_DNS_Record_Types_AWS** lab to host the website using EC2.
- Ensure two EC2 web servers are running in different Availability Zones.



- Verify that both web servers host the same website.

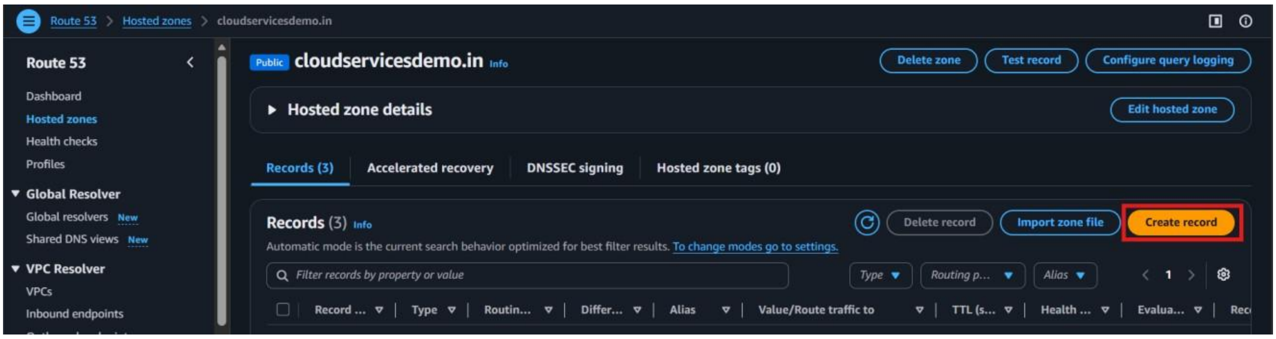


- Note the public IPv4 address of each web server.



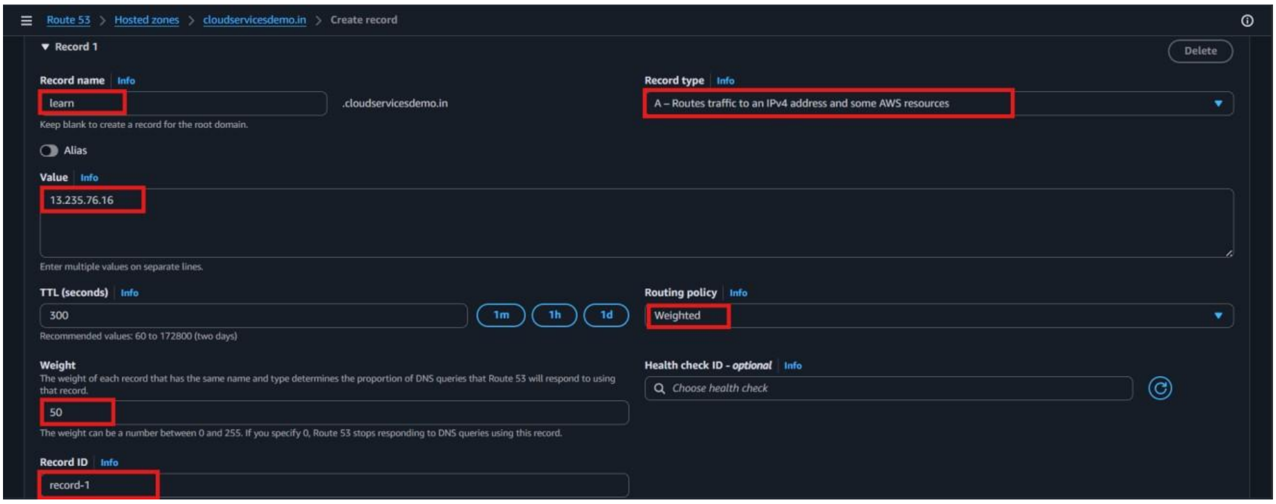
Step 2 – Create the First Weighted Record

- Open the **Amazon Route 53** console.
- Select the existing **Public Hosted Zone**.
- Click **Create Record**.



- Enter the following:
    - **Record Name:** learn

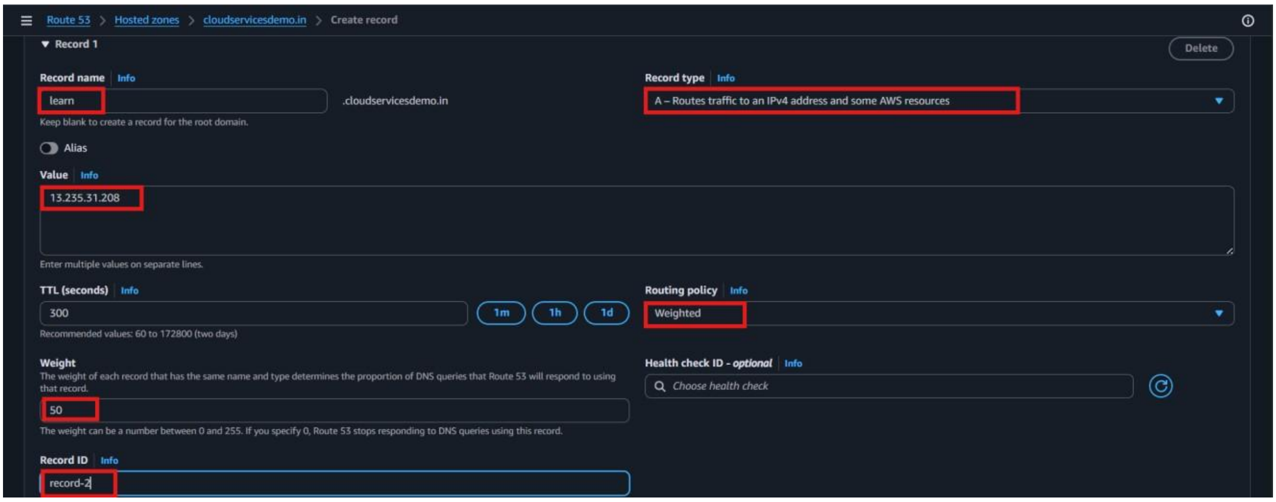
- Record Type: A
  - **Value:** (Public IPv4 address of web-Server.)
  - **Routing Policy:** Weighted
  - **Weight:** 50
  - **Record ID:** record-1



- Click **Create Records**.

Step 3 – Create the Second Weighted Record

- Click **Create Record**.
- Enter the following:
  - **Record Name:** learn
  - **Record Type:** A
  - **Value:** Public IPv4 address of Web Server 2.
  - **Routing Policy:** Weighted
  - **Weight:** 50
  - **Record ID:** record-2



- Click **Create Records**.
- Verify that two records with the same name are created successfully.

Step 4 – Test the Weighted Routing Policy

- Open the **Route 53** console.
- Select the hosted zone.
- Query the record multiple times.
- Verify that the returned IP address alternates between both web servers.
- Alternatively, open **Command Prompt** and run:
    - **nslookup** learn.<**your-domain-name**>
- Run the command multiple times.

```
PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address:  13.235.31.208

PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address:  13.235.31.208

PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address:  13.235.76.16

PS C:\Users\HP\downloads>
```

- Verify that the response resolves to both IP addresses.

Step 5 – Understand Weight Distribution

- Review how Route 53 distributes traffic based on the assigned weights.
- **Run**:
  1..10 | ForEach-Object { nslookup learn.cloudservicesdemo.in 8.8.8.8 }
- Observe that:
    - Equal weights distribute traffic equally.
    - Higher weights receive a larger percentage of traffic.
- Use the following formula to calculate traffic distribution:

  Traffic Percentage = (Assigned Weight ÷ Total Weight) × 100

Step 6 – Understand Route 53 Health Checks

- Review how Route 53 Health Checks monitor the availability of web servers.
- Understand that if one server becomes unhealthy:

  - Route 53 stops returning its IP address.
  - Traffic is routed only to healthy servers.
- When the failed server becomes healthy again, Route 53 resumes traffic distribution according to the configured weights.

Verification

- Configured a Weighted Routing Policy successfully.
- Created multiple weighted A records for the same domain name.
- Verified DNS traffic distribution across multiple web servers.
- Understood how Route 53 calculates weighted traffic distribution.
- Reviewed how Route 53 Health Checks improve application availability.